WS-2026 ADVANCED HARDWARE ENGINEERING · HOCHSCHULE HAMM-LIPPSTADT

Lab 03 – MicroBlaze with AXI GPIO, 7-Segment Display and Own PWM IP

Final summary · Nexys A7-100T (xc7a100tcsg324-1) · Vivado 2025.2 / Vitis Unified IDE 2025.2

Result	Value
Demo	Switches control LEDs, 7-segment display and RGB LED ✔
Timing at 100 MHz	Met – WNS 1.472 ns, WHS 0.045 ns, 0 failing endpoints
Utilization	1,795 LUTs (2.8 %) · 2,019 FFs (1.6 %) · 32 BRAM (23.7 %) · 55 IOB
Power	0.236 W total on-chip
Software	gpio_demo.elf, 11,392 bytes of 128 KB local memory

1. Objective

Lab 03 extends the MicroBlaze system from Lab 02 with memory-mapped peripherals and connects them to the board I/O. Two catalog **AXI GPIO** blocks drive the 16 switches, 16 LEDs and the 8-digit 7-segment display, and the custom **pwm_axi** IP from Lab 01 drives the RGB LED LD16. A single C program reads the switches and updates all three outputs in one loop.

Reused from earlier labs:

- **Lab 02:** complete MicroBlaze system (MicroBlaze, 128 KB LMB BRAM, Clocking Wizard, Processor System Reset, AXI UART Lite at 9600 baud, interrupt controller, MDM), copied with *Save Project As*.
- **Lab 01:** the packaged AXI4-Lite IP **pwm_axi v1.0** (8-bit PWM core, 3 channels) from *ip_repo*, plus the verified switch and LED pin assignments.

2. Hardware design

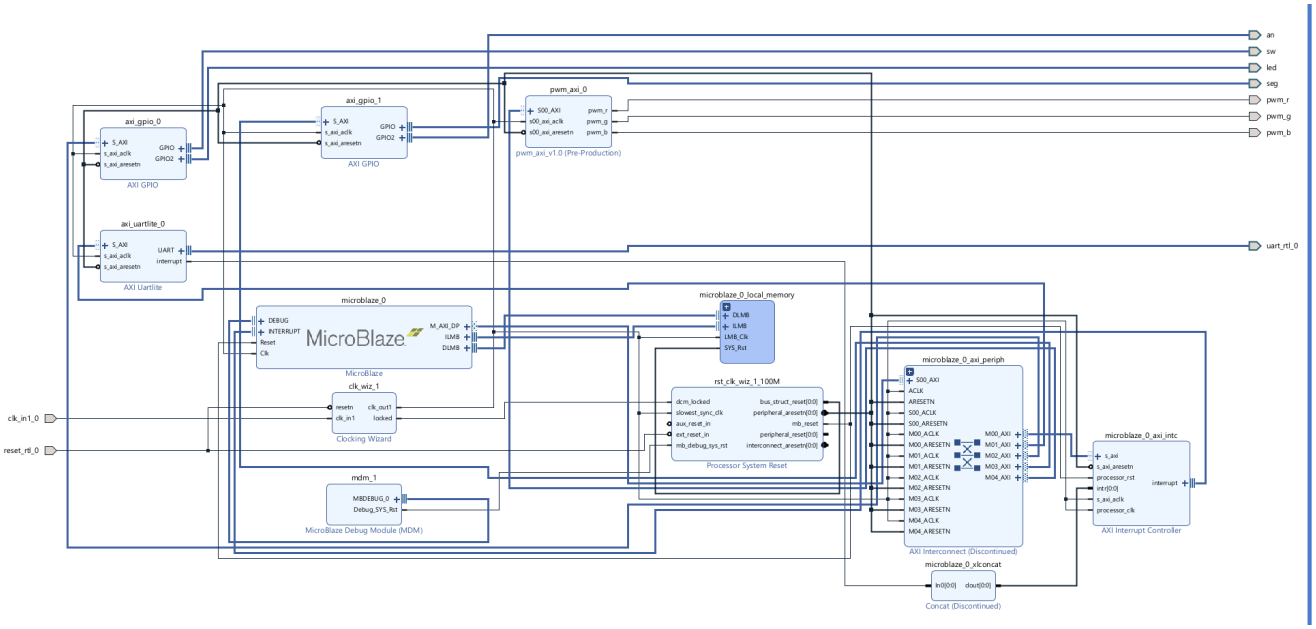


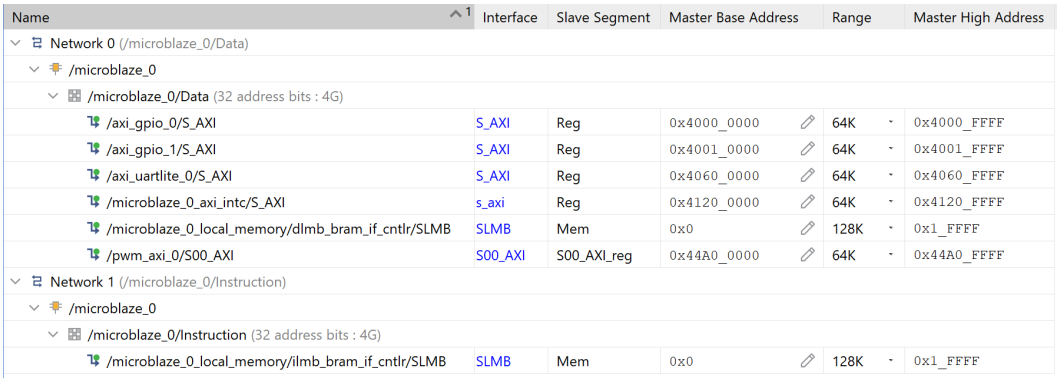
Figure 1 – Block design with axi_gpio_0, axi_gpio_1 and pwm_axi_0 connected to the MicroBlaze through the AXI interconnect (M00–M04).

New blocks and configuration

Block	Configuration	External port → board
axi_gpio_0	Dual channel. GPIO: 16 bit, all inputs. GPIO2: 16 bit, all outputs, default 0x0	sw → SW0–SW15, led → LD0–LD15
axi_gpio_1	Dual channel. GPIO and GPIO2: 8 bit, all outputs, default 0xFF (display off)	seg → CA–CG, DP; an → AN0–AN7
pwm_axi_0	Own IP from Lab 01, AXI4-Lite, 4 × 32-bit registers	pwm_r/g/b → LD16 (N15, M16, R12)
AXI interconnect	Extended from 2 to 5 master ports by Connection Automation	–

Address map

Peripheral	Base address	High address	Range	Use in software
axi_gpio_0	0x4000_0000	0x4000_FFFF	64 K	Switches (0x00), LEDs (0x08)
axi_gpio_1	0x4001_0000	0x4001_FFFF	64 K	Segments (0x00), digit enables (0x08)
axi_uartlite_0	0x4060_0000	0x4060_FFFF	64 K	xil_printf (from Lab 02)
axi_intc	0x4120_0000	0x4120_FFFF	64 K	Not used (from Lab 02)
pwm_axi_0	0x44A0_0000	0x44A0_FFFF	64 K	CTRL, R, G, B duty
Local memory (I/D)	0x0000_0000	0x0001_FFFF	128 K	Code and data



Name	Interface	Slave Segment	Master Base Address	Range	Master High Address
Network 0 (/microblaze_0/Data)					
/microblaze_0					
/microblaze_0/Data (32 address bits : 4G)					
/axi_gpio_0/S_AXI	S_AXI	Reg	0x4000_0000	64K	0x4000_FFFF
/axi_gpio_1/S_AXI	S_AXI	Reg	0x4001_0000	64K	0x4001_FFFF
/axi_uartlite_0/S_AXI	S_AXI	Reg	0x4060_0000	64K	0x4060_FFFF
/microblaze_0_axi_intc/S_AXI	s_axi	Reg	0x4120_0000	64K	0x4120_FFFF
/microblaze_0_local_memory/dlmb_bram_if_cntlr/SLMB	SLMB	Mem	0x0	128K	0x1_FFFF
/pwm_axi_0/S00_AXI	S00_AXI	S00_AXI_reg	0x44A0_0000	64K	0x44A0_FFFF
Network 1 (/microblaze_0/Instruction)					
/microblaze_0					
/microblaze_0/Instruction (32 address bits : 4G)					
/microblaze_0_local_memory/ilmb_bram_if_cntlr/SLMB	SLMB	Mem	0x0	128K	0x1_FFFF

Figure 2 – Address Editor in Vivado.

Constraints (nexys_a7_pins.xdc)

- 55 user I/O: clock, reset and UART from Lab 02, plus 16 switches, 16 LEDs, 8 segments, 8 anodes and 3 PWM outputs.
- Port names follow the wrapper generated by *Make External*: `sw_tri_i`, `led_tri_o`, `seg_tri_o`, `an_tri_o`, `pwm_r/g/b`.
- SW8 (T8) and SW9 (U8) use **LVCMOS18** because they sit in a 1.8 V bank (taken from Lab 01); all other pins LVCMOS33.
- The duplicate `create_clock` from Lab 02 was removed (the Clocking Wizard already constrains clk_in1_0), which clears warning XDCC-7.
- `CFGBVS VCCO` / `CONFIG_VOLTAGE 3.3` added, which clears DRC warning CFGBVS-1.

3. Software

The application **gpio_demo** runs bare-metal (standalone BSP) on microblaze_0. All peripherals are accessed as memory-mapped registers with `Xil_In32` / `Xil_Out32` at *base address* + *offset*. The source file is fully commented at every register access.

Register maps used

Peripheral	Offset	Register	Bits used / meaning
axi_gpio_0	0x00	GPIO_DATA (ch 1)	[15:0] switch states, read
axi_gpio_0	0x08	GPIO2_DATA (ch 2)	[15:0] LED states, write (1 = on)
axi_gpio_1	0x00 / 0x04	GPIO_DATA / GPIO_TRI	[7:0] segments CA–CG, DP (active low) / direction 0 = output
axi_gpio_1	0x08 / 0x0C	GPIO2_DATA / GPIO2_TRI	[7:0] digit enables AN0–AN7 (active low) / direction
pwm_axi_0	0x00	CTRL (slv_reg0)	bit 0 = enable for all three channels
pwm_axi_0	0x04 / 0x08 / 0x0C	R / G / B (slv_reg1–3)	[7:0] duty 0–255

Program flow

- Initialisation: GPIO1 direction = output, display blanked (0xFF), all PWM duties 0, then CTRL = 1. A start-up message is printed over the UART.
- Main loop: read the 16 switches → write them to the LEDs → write the three PWM duties → show the switch value in hexadecimal on the four right-hand digits.
- Colour mapping: SW3..0 = red, SW7..4 = green, SW11..8 = blue. Each 4-bit level 0–15 is multiplied by 17 to cover the duty range 0–255.
- Display: `show_hex()` multiplexes four digits. For each digit it blanks all anodes, writes the segment pattern from a 16-entry table, enables one anode and waits 1 ms.

```
while (1) {
    u32 sw = Xil_In32(GPIO0_BASE + GPIO_SW) & 0xFFFF;   /* read switches   */
    Xil_Out32(GPIO0_BASE + GPIO_LED, sw);                /* LEDs = switches */
    Xil_Out32(PWM_BASE + PWM_R, ((sw >> 0) & 0xF) * 17); /* SW3..0  red     */
    Xil_Out32(PWM_BASE + PWM_G, ((sw >> 4) & 0xF) * 17); /* SW7..4  green   */
    Xil_Out32(PWM_BASE + PWM_B, ((sw >> 8) & 0xF) * 17); /* SW11..8 blue    */
    show_hex((u16)sw);                                   /* 4 digits, hex   */
}
```

Build result: **text 7,516 B, data 412 B, bss 3,464 B – total 11,392 B**, about 9 % of the 128 KB local memory. The editor's clangd hints (unknown argument `-mxl-soft-mul`, missing `xpseudo_asm.h`) are code-indexer warnings only; the MicroBlaze GCC build finished without errors.

4. Implementation results and comparison with Lab 02

Resource (available)	Lab 02	Lab 03	Change
Slice LUTs (63,400)	1,515 (2.39 %)	1,795 (2.83 %)	+280 (+18 %)
LUT as logic	1,345	1,625	+280
LUT as memory (19,000)	170	170	0
Slice registers (126,800)	1,538 (1.21 %)	2,019 (1.59 %)	+481 (+31 %)
F7 muxes (31,700)	109	109	0
Slices (15,850)	573 (3.62 %)	713 (4.50 %)	+140
Block RAM tiles (135)	32 (23.7 %)	32 (23.7 %)	0
DSP48 (240)	0	0	0
Bonded IOB (210)	4 (1.9 %)	55 (26.2 %)	+51
BUFGCTRL / MMCM	4 / 1	4 / 1	0

Name	Slice LUTs (63400)	Slice Registers (126800)	F7 Muxes (31700)	Slice (15850)	LUT as Logic (63400)	LUT as Memory (19000)	Block RAM Tile (135)	Bonded IOB (210)
design_microblaze_wrapper	1795	2019	109	713	1625	170	32	55
design_microblaze_i (design_microblaze)	1795	2019	109	713	1625	170	32	0

Figure 3 – Utilization of the implemented Lab 03 design (Vivado, hierarchy view).

Where the additional logic comes from

Out-of-context synthesis results of the new or changed blocks (before global optimisation, so the sum is slightly above the final difference):

Block	LUTs	Registers	Comment
pwm_axi_0	86	153	Same as the standalone Lab 01 result; 128 FFs are the four 32-bit AXI registers
axi_gpio_0 (16 in + 16 out)	66	272	Data, direction and input-synchroniser registers
axi_gpio_1 (8 out + 8 out)	42	144	Smaller because both channels are 8 bit
AXI crossbar (2 → 5 slaves)	119 → 227	130 → 134	+108 LUTs for address decoding of 3 more slaves

- **I/O +51** matches exactly 16 switches + 16 LEDs + 8 segments + 8 anodes + 3 PWM outputs.
- **BRAM, LUT-RAM and F7 muxes are unchanged**: the processor and its 128 KB memory are identical to Lab 02; all growth comes from the new peripherals and the larger interconnect.
- The whole system still uses less than 5 % of the slices of the XC7A100T.

Timing and power

Metric	Lab 02	Lab 03	Comment
WNS (setup)	1.956 ns	1.472 ns	Met; $f_{max} \approx 1 / (10 - 1.472)$ ns ≈ 117 MHz
WHS (hold)	0.047 ns	0.045 ns	Met
Timing endpoints	5,892	6,999	More registers in the new peripherals
Total on-chip power	0.230 W	0.236 W	Dynamic 0.131 → 0.137 W, static 0.099 W

Setup

Worst Negative Slack (WNS):	1.472 ns
Total Negative Slack (TNS):	0.000 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	6999

Hold

Worst Hold Slack (WHS):	0.045 ns
Total Hold Slack (THS):	0.000 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	6999

Pulse Width

Worst Pulse Width Slack (WPWS):	3.000 ns
Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	2401

All user specified timing constraints are met.

Figure 4 – Design Timing Summary: all user-specified timing constraints are met.

The setup slack is lower than in Lab 02 because the critical path now runs through the larger 5-port crossbar, but the 100 MHz system clock is still met with margin. Power hardly changes: the MMCM and clock tree dominate, and GPIO and PWM logic toggles very little.

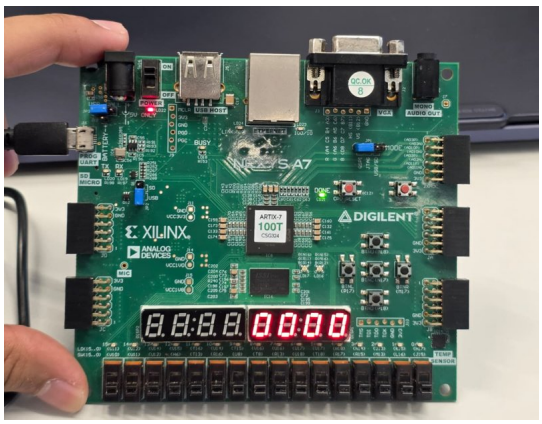
Warnings

Check	Lab 02	Lab 03
XDCC-7 (clock constraint overwritten)	1	0 – duplicate create_clock removed
CFGBVS-1 (DRC)	1	0 – CFGBVS / CONFIG_VOLTAGE set
TIMING-18 (missing I/O delay)	3	54 – one per slow board I/O; acceptable for switches and LEDs
LUTAR-1, TIMING-9	1 + 1	1 + 1 – inside the reset / debug IP, unchanged

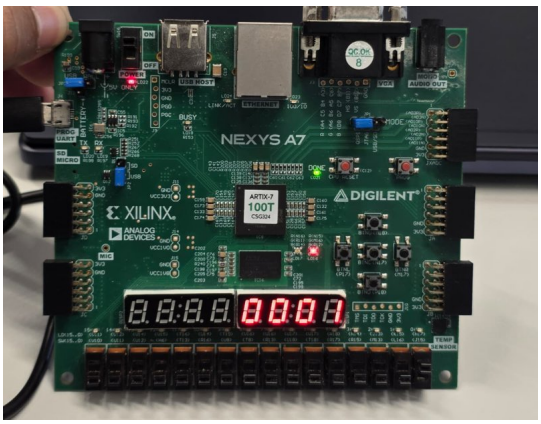
5. Demonstration on the Nexys A7

The FPGA was programmed from Vitis with the bitstream included in the XSA, and gpio_demo.elf was run on microblaze_0. The serial monitor (COM6, 9600 baud) showed the start-up message, and all outputs reacted to the switches:

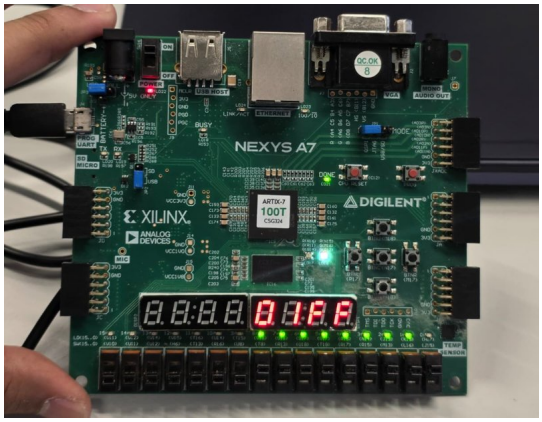
Switches up	Display	LEDs	RGB LED LD16	Shows
none	0000	all off	off	Initial state, PWM duty 0
SW0	0001	LD0	dim red (duty 17/255)	PWM duty from software
SW0–SW8	01FF	LD0–LD8	R = G = 255, B = 17	Colour mixing of three channels
all 16	FFFF	all on	R = G = B = 255 (white)	Full range of GPIO and PWM



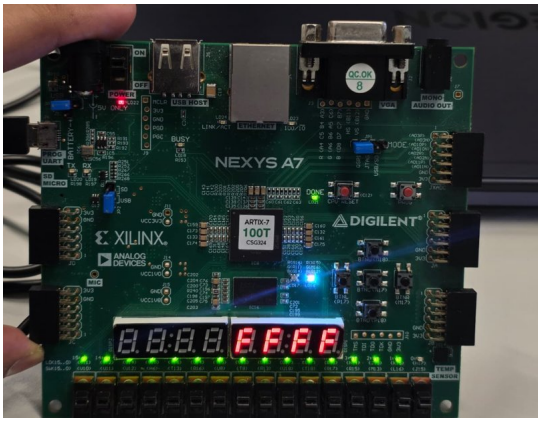
0000 – all switches off



0001 – SW0, dim red



01FF – SW0–SW8



FFFF – all switches on

Figure 5 – Board photos. The four right-hand digits show the switch value in hexadecimal; the left four digits are not driven by the program and stay dark.

6. Answers to the questions in the lab

Which bits of the PWM registers are used?

Only bit 0 of CTRL (one enable shared by R, G and B) and bits [7:0] of the three duty registers. The upper bits are stored by the AXI slave and can be read back, but the PWM core ignores them.

What happens at duty 0 and 255?

Duty 0 keeps the output permanently low (LED off). Duty 255 gives 255/256 = 99.6 % high time, never fully 100 %, because the output is high only while the 8-bit counter is smaller than the duty value. The PWM frequency is 100 MHz / 256 = 390.6 kHz, far above visible flicker. The Lab 01 testbench showed the same behaviour at 4-bit width (duty 15 → 15/16 cycles high).

Why are all segments lit before the program runs?

Segments and anodes are active low. After configuration the AXI GPIO outputs take their default value 0x0, which switches every segment and digit on. Setting the default output value of axi_gpio_1 to 0xFF and blanking the display at the start of main() keeps it dark until the program writes valid data.

Why does the display need multiplexing, and what does the delay change?

All eight digits share the same eight segment lines, so only one digit can show its own pattern at a time. With 1 ms per digit, one pass over four digits takes about 4 ms (≈ 250 Hz per digit), so the eye sees a steady display. A longer delay such as 5 ms gives 20 ms per pass (50 Hz) and visible flicker; a very short delay removes flicker but makes the digits dimmer, because switching overhead takes a larger share of each slot. usleep() is a calibrated busy-wait here, since the design has no hardware timer.

Why must the new bitstream be loaded before the program?

An old bitstream does not contain the GPIO and PWM peripherals, so the first access to their addresses never gets an AXI response and the processor hangs. The XSA was therefore exported with the bitstream included, so Vitis programs the matching hardware before it starts the ELF.

7. Problems encountered and solutions

Problem	Cause	Solution
IP locked after opening Lab 02	Project created in Vivado 2025.1, opened in 2025.2	Report IP Status → Upgrade Selected (revision changes only)
Failed to save interconnect XCI	Windows 260-character path limit in a deep folder	Save Project As into the short path C:\AHE
Synthesis: top module not found	XDC text pasted into the wrapper VHDL by mistake	Wrapper regenerated, constraints pasted into nexys_a7_pins.xdc
Vitis offered only HLS components	Only Vivado ML Enterprise installed; marker file .vitis_for_hls	Added Vitis Unified Software Platform; marker file renamed
Link error: undefined reference to main	main.c was empty after import / moved from its build path	Kept main.c in its original location and pasted the full source

8. Summary

Memory-mapped I/O. Every peripheral appears to the MicroBlaze as a block of registers at a fixed address from the Address Editor. Software controls hardware with plain 32-bit reads and writes to base + offset; no special instructions are needed.

AXI GPIO. The catalog GPIO provides up to two channels per instance, each with a data and a direction register. Two instances cover four pin groups: switches (input), LEDs, segments and anodes (outputs). Fixed "all inputs / all outputs" configuration removes the need to program the direction registers.

Multiplexed display. Eight digits share eight segment lines; software enables one active-low anode at a time and refreshes each digit about 250 times per second, so all digits appear lit while only 16 pins are used.

Reuse of own IP. The pwm_axi IP from Lab 01 was added from the IP repository and wired by Connection Automation exactly like a catalog IP. Its documented AXI4-Lite register map is all the software needs. It cost 86 LUTs and 153 registers, and the complete system still uses under 3 % of the LUTs while meeting 100 MHz timing.